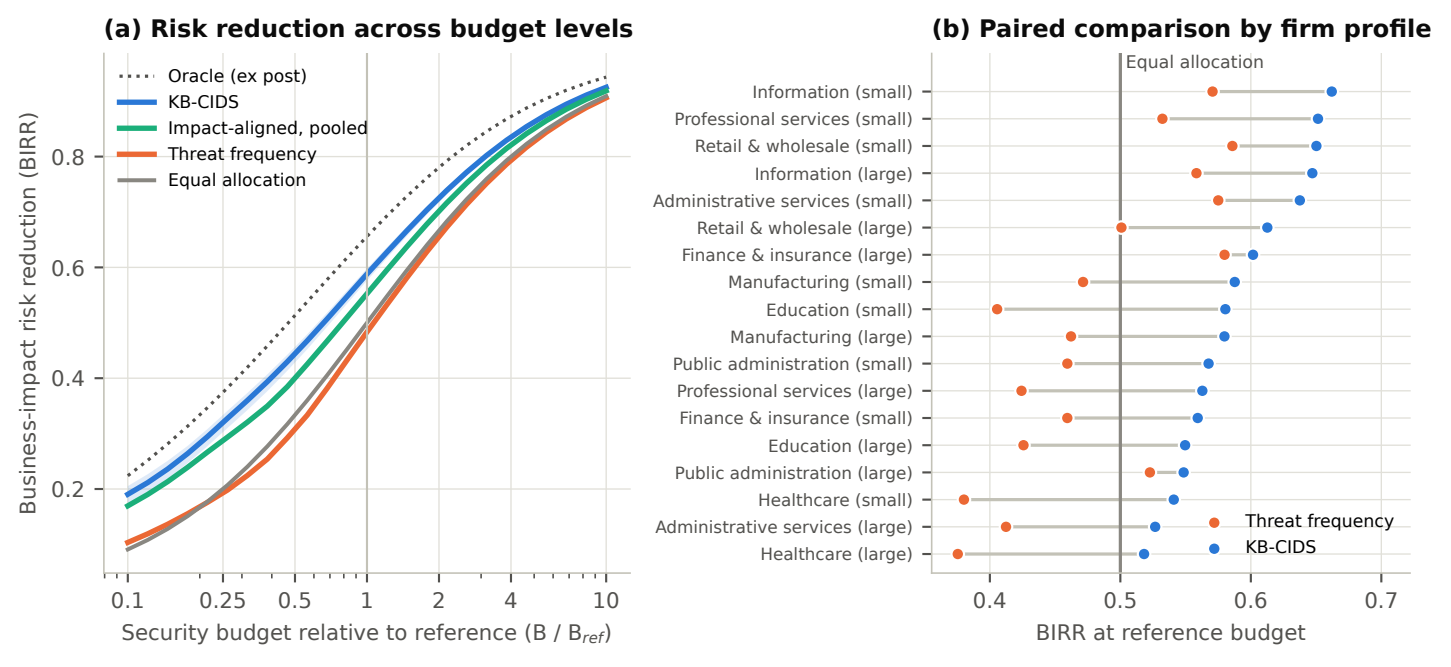
(a) Risk reduction across budget levels
Oracle (ex post)
KB-CIDS
Impact-aligned, pooled
Threat frequency
Equal allocation
Business-impact risk reduction (BIRR)
0.2
0.4
0.6
0.8
0.1
0.25
0.5
1
2
4
10
Security budget relative to reference (B / B$_{ref}$)
(b) Paired comparison by firm profile
Equal allocation
Information (small)
Professional services (small)
Retail & wholesale (small)
Information (large)
Administrative services (small)
Retail & wholesale (large)
Finance & insurance (large)
Manufacturing (small)
Education (small)
Manufacturing (large)
Public administration (small)
Professional services (large)
Finance & insurance (small)
Education (large)
Public administration (large)
Healthcare (small)
Administrative services (large)
Healthcare (large)
Threat frequency
KB-CIDS
0.4
0.5
0.6
0.7
BIRR at reference budget